INFRASTRUCTURE

Digital twins are being deployed in critical infrastructure projects, from individual facilities to entire regional networks. Power systems use digital twins to balance supply and demand in real-time, integrating data from generation facilities, transmission lines, and smart meters to prevent outages. The Netherlands employs a digital twin of its flood protection infrastructure to simulate storm scenarios and optimize barrier operations protecting millions of residents below sea level. Digital twins enable infrastructure operators to orchestrate entire systems rather than just individual assets, ensuring the reliability and resilience that modern society depends upon.

SMART BUILDINGS, CITIES, AND STATES

Digital twins are being used across multiple scales in the built environment, from individual building systems to entire cities and states. Within individual buildings, digital twins are used to optimize energy consumption against the competing priorities of energy use and occupant comfort by integrating data from occupancy, weather, and other types of sensors. Saudi Arabia, with an ambitious plan to house 9 million people in a footprint of 34 square kilometers, uses a digital twin to simulate the operation of lifts within high-rise buildings to optimize the design and construction process.

1.5 Building your first digital twin: key considerations and challenges

You might be wondering where you would even start building a digital twin. Whatever level of digital twin you are planning, the components of the system are as shown in figure 1.12.